

# Maze Solver Robot using Digital Systems

MINI PROJECT REPORT

*submitted by*

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*in partial fulfillment for the completion of  
EI23402 DIGITAL SYSTEM LAB*



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## **BONAFIDE CERTIFICATE**

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# Maze Solver Robot using Digital Systems

## Introduction

This mini project presents a simple **Maze Solver Robot** designed using the principles of **Digital Systems**. It uses **two IR sensors** – one placed **in front** and another on the **right side** of the robot – to detect obstacles.

Using **NOT gates** and **Boolean logic**, the robot decides movement directions based on obstacle presence. The movement is controlled via an **L293D motor driver**, enabling the robot to smartly navigate a maze.

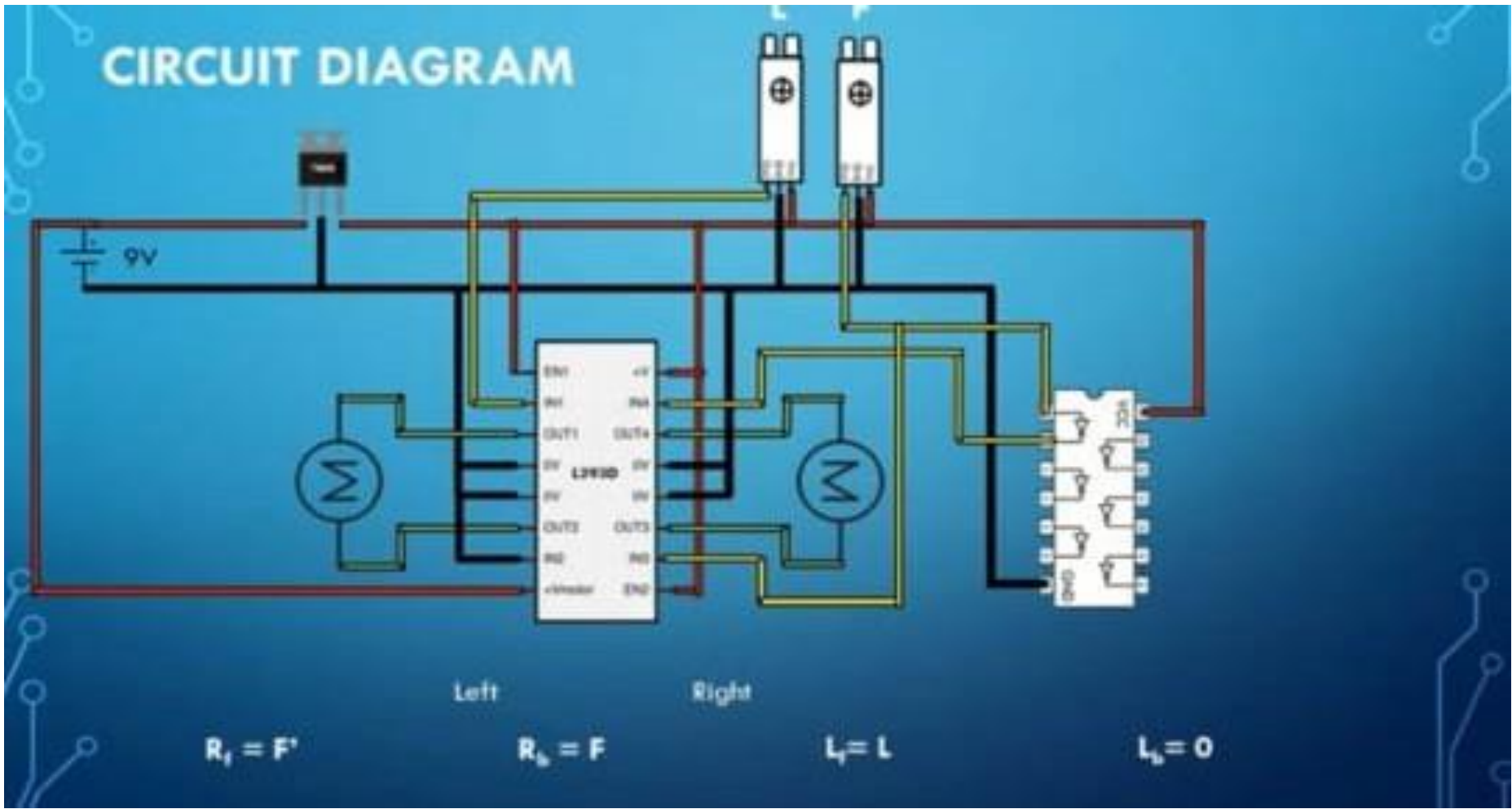
This project highlights the practical use of **Boolean algebra**, **truth tables**, and **combinational logic** for real-world problem solving.

## Components Used

| S.No | Component                  | Quantity |
|------|----------------------------|----------|
| 1    | IR Obstacle Sensor (Front) | 1        |
| 2    | IR Obstacle Sensor (Right) | 1        |
| 3    | DC Motors                  | 2        |
| 4    | L293D Motor Driver IC      | 1        |
| 5    | NOT Gates (7404 IC         | 1 IC     |

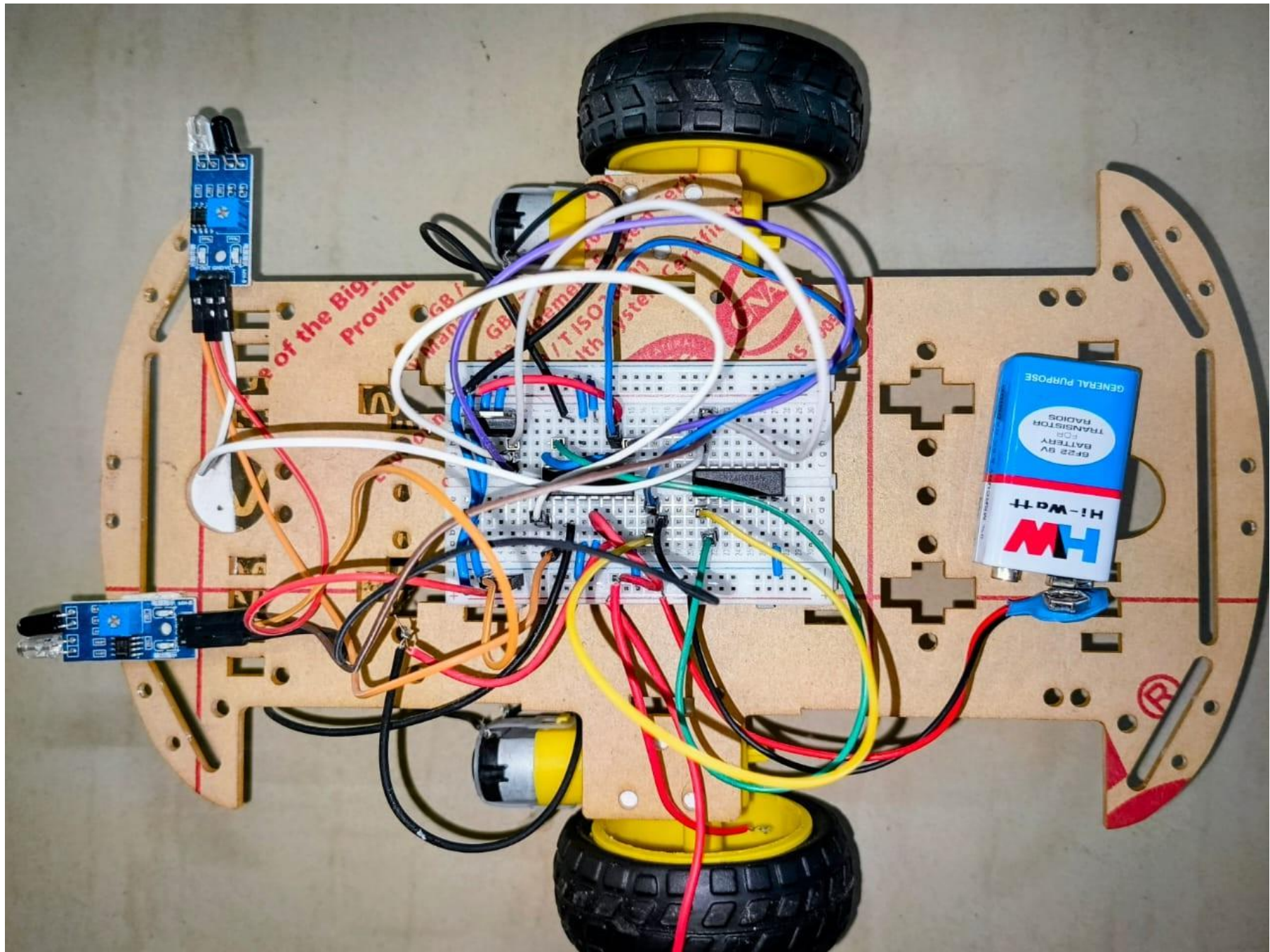
| S.No | Component               | Quantity    |
|------|-------------------------|-------------|
|      | or similar)             |             |
| 6    | Arduino / Logic Circuit | 1           |
| 7    | 9V Battery              | 1           |
| 8    | Robot Chassis           | 1           |
| 9    | Connecting Wires        | As required |

**CIRCUIT DIAGRAM:**



**HARDWARE IMPLEMENTATION:**





### Truth Table

| Front (F) | Right (R) | F' | R' | Rf | Rb | Lf | Lb | Action             |
|-----------|-----------|----|----|----|----|----|----|--------------------|
| 0         | 0         | 1  | 1  | 1  | 0  | 1  | 0  | Move Forward       |
| 0         | 1         | 1  | 0  | 1  | 0  | 0  | 0  | Slight Left Turn   |
| 1         | 0         | 0  | 1  | 0  | 1  | 1  | 0  | Turn Right         |
| 1         | 1         | 0  | 0  | 0  | 1  | 0  | 0  | Move Backward&Left |

### Working of the Maze Solver Robot

The Maze Solver Robot is designed to autonomously navigate a maze using two IR sensors:

- ✦ One placed at the front of the robot to detect obstacles ahead, and
- ✦ One placed on the right side to detect obstacles to the right.

The robot uses NOT gates and Boolean logic to determine the direction of movement based on sensor readings.

Step-by-Step Operation:

1. Initial Condition:
  - The robot continuously reads the outputs from the Front IR sensor (F) and Right IR sensor (R).
  - These outputs are digital:  
0 = No obstacle, 1 = Obstacle detected.
- 2 Logic Evaluation:
  - The sensor outputs are passed through NOT gates to produce:  
F' (inverted front sensor): 1 means no obstacle ahead.  
R' (inverted right sensor): 1 means no obstacle on the right.
  - These values are used in Boolean expressions to control motor movement.
- 3 Motor Control Logic: Based on sensor inputs, the robot takes one of the following actions:

| F<br>(Front) | R<br>(Right) | Action Taken                                            |
|--------------|--------------|---------------------------------------------------------|
| 0            | 0            | Move Forward: No obstacles, both motors ON.             |
| 0            | 1            | Turn Slightly Left: Stop left motor, right motor moves. |



| F<br>(Front) | R<br>(Right) | Action Taken                                              |
|--------------|--------------|-----------------------------------------------------------|
| 1            | 0            | Turn Right: Obstacle ahead, turn robot right.             |
| 1            | 1            | Move Backward + Turn Left: Obstacle both front and right. |

#### 4. Motor Activation (Using L293D):

- The L293D motor driver receives logic signals from the control circuit.
- It activates motors as per the Boolean output:

$$R_f = F'$$

$$R_b = F$$

$$L_f = R'$$

$$L_b = 0 \text{ (Always OFF)}$$

#### 5. Navigation Result:

- The robot follows the maze path by:

Moving forward when clear.

Slightly turning left when there's an obstacle on the right.

Turning right when the front is blocked but the right is free.

Moving backward and rotating left if blocked both front and right.

### Advantages

- † Easy to implement with only **combinational logic**

- ‡ Efficient decision-making using simple logic gates
- ‡ Educational: Demonstrates **real-life Boolean logic**
- ‡ Avoids complex microcontroller coding
- ‡ Expandable to 3- or 4-sensor designs

## Limitations

- ‡ Limited to front and right obstacle detection only
- ‡ Not suitable for dynamic or complex environments
- ‡ IR sensors have reduced accuracy on shiny or dark surfaces
- ‡ No memory or learning – purely reactive system

## Applications

- ‡ Intro to robotics and sensor integration
- ‡ Basic autonomous robot design
- ‡ Maze-solving contests or educational demos

## Conclusion

This project applies core **Digital Systems concepts** – such as **truth tables**, **Boolean expressions**, and **NOT gates** – to create an intelligent robot that solves mazes using only **two sensors and combinational logic**.

The logic is simple but effective. Using just two NOT gates and a motor driver, this robot demonstrates how real-world systems can be controlled using basic logic – a perfect example of theory meeting practical a